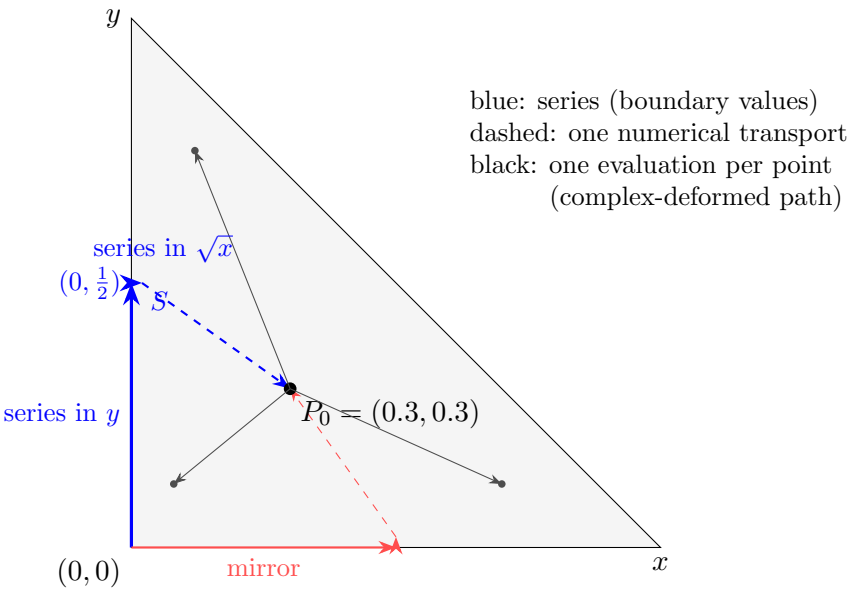
Numerical evaluation of the $V \to q\bar{q}g$ finite remainders: boundary values and transport

Setup. Every finite remainder is a sum $\sum q\, R_r(y,z)\, T_g\, c_m$ of rational functions, constants and 579 pure functions T_g. The T_g are iterated integrals with base point $(x,y) = (0,0)$, where every non-empty integral vanishes after shuffle regularisation. They satisfy a closed canonical system $dF = \sum_k d\log W_k\, M_k F$ of 503 functions, which we integrate numerically.

Boundary values (computed once). The origin is singular, so we do not start there. We expand in series instead (see the figure):

1. along the edge $x = 0$, from $(0,0)$ to $(0, \frac{1}{2})$, as a series in y. On the edge only $y = 0, 1$ are singular, and the regularisation sets $\log y \to 0$ at the origin;
2. off the edge at fixed $y = \frac{1}{2}$, as a series in $\sqrt{x}$ (the square-root letters go like $\sqrt{x}$), up to $S = (0.02, \frac{1}{2})$. Here $\log x \to 0$ at $x = 0$;
3. one numerical transport (Bulirsch–Stoer) from S to the central point $P_0 = (0.3, 0.3)$.

The values at P_0 are stored once and for all and shipped with the evaluator. The mirror construction, through $(\frac{1}{2}, 0)$ and $(\frac{1}{2}, 0.02)$, gives an independent set, used only for the error estimate.



Evaluation at a point P. We integrate the system from P_0 to P along a straight path, deformed into the complex plane ($z(t) = P_0 + (t + 4i\delta\, t(1-t))(P - P_0)$). Three refinements:

- *Spurious letters.* Six letters vanish on curves inside the triangle. The amplitude is smooth there, but the equations are not. The deformation δ is chosen so that the complex path stays away from their zeros.
- *Points on a spurious curve.* Such a point is replaced by the symmetric average of $P \pm h\,n$, $P \pm 2h\,n$, $P \pm 3h\,n$ across the curve (n its normal), which is accurate to $O(h^6)$.
- *Error estimate.* Each point is evaluated a second time, from the mirror boundary values and on another path; the estimate is ten times the difference.

In the bulk the accuracy is about 10^{-12}, and about 10^{-13} against the exact planar results. Near the edges double precision loses digits, which the error estimate reports; there the corner expansions take over.

High-precision values at P_0 (27 digits)

1. A double series at the corner. Here we use the exact system: 586 functions in an integral basis, with integer matrices M_k (`out/basis.wl`, `out/deq.wl`); the 579 T_g are integer combinations of them. Near $(0,0)$ every function is a double series

$$F(x,y) = \sum c\, x^{a/2}\, y^{b/2}\, \log^k x\, \log^l y, \qquad c \text{ exact rational (times 1 or } i).$$

We obtain it from the blow-up series of the ITER family: chart 1 ($x = t\,y^2$) and its mirror chart 4 ($y = t\,x^2$), each to order t^{16} and B^{48} (B the other variable). We undo the blow-up: $t = x/y^2$, $\log t = \log x - 2\log y$.

- No negative powers appear, and the two charts agree exactly on all $1.3 \cdot 10^6$ terms they share. Several letters vanish at the corner ($x+y$, $x-y+y^2$, $(x+y)^2 - x$, …), but the functions do not see them there: the blow-up is needed for the rational functions only.
- The union of the two charts holds every term of total degree $\frac{a+b}{2} \le 32$ (the first missing one is $x^{16}y^{17}$). The truncation error is about $(|x|+|y|)^{33}$: 10^{-34} at $(0.05, 0.05)$, 10^{-27} at $(0.08, 0.08)$.

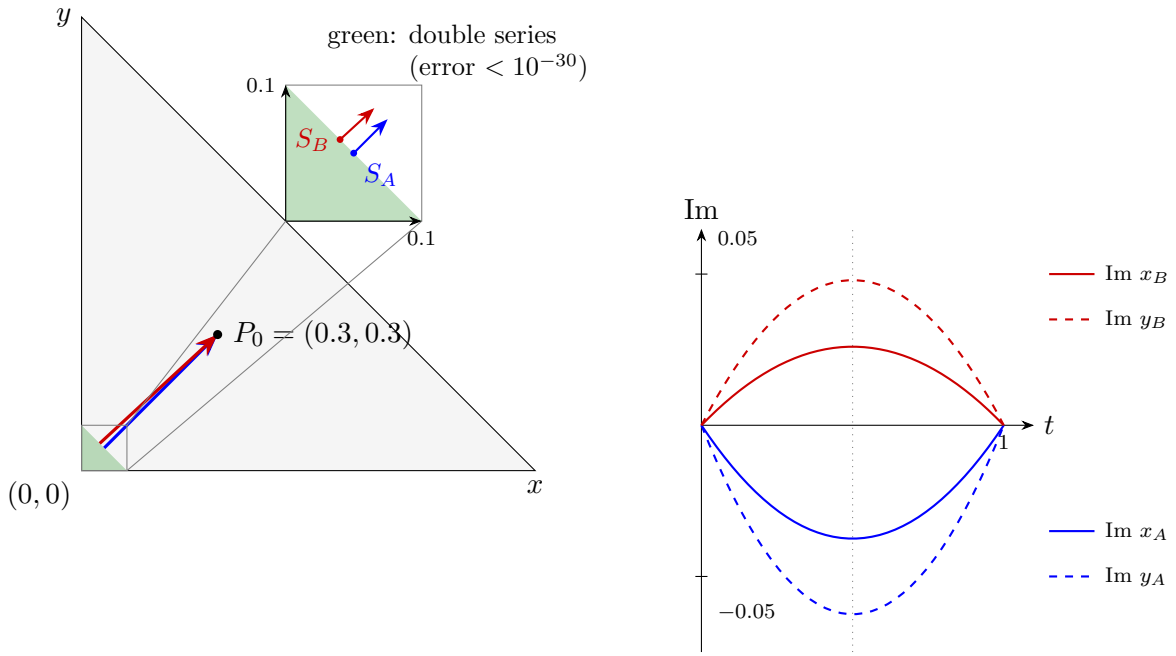
2. Two start points, evaluated from the series at 40 digits:

$$S_A = (0.05,\ 0.05), \qquad S_B = (0.04,\ 0.06).$$

3. Transport to P_0 in quad precision. From each start point we integrate to $P_0 = (0.3, 0.3)$ along its own complex-deformed straight path,

$$x(t) = S_x + \big(t + 4i\,\delta_x\, t(1-t)\big)(0.3 - S_x), \qquad y(t) = S_y + \big(t + 4i\,\delta_y\, t(1-t)\big)(0.3 - S_y), \qquad 0 \le t \le 1,$$

with $\delta_A = (-0.15,\ -0.25)$ (below the real plane) and $\delta_B = (0.1,\ 0.2)$ (above). The root $r = \sqrt{xyz}$ is integrated along ($r > 0$ at S). The integrator is Bulirsch–Stoer (modified midpoint, up to 16 extrapolation levels) in double-double arithmetic (≈ 32 digits), with tolerance 10^{-32} relative to the largest function. Each path takes about one minute (`cpp/iter_quad`, `quad_data.py`).



Left: the real part of the two paths (straight lines), with a zoom of the corner. Right: their imaginary parts along the path, $4\delta\, t(1-t)(0.3-S)$; S_A runs below the real plane, S_B above.

4. Result. The two determinations agree to $3 \cdot 10^{-28}$ relative to the largest function ($4.5 \cdot 10^4$), i.e. 27–28 digits. Further checks, all within $2 \cdot 10^{-27}$: a second path from S_A ($\delta = (0.1, 0.2)$), and start points $(0.06, 0.06)$, $(0.06, 0.04)$, $(0.08, 0.08)$; the last differs by its series truncation, $2 \cdot 10^{-27}$. The average is stored in `out/iter_quad_P0.bin` (input of `cpp/iter_quad`) and `out/P0_values_quad.m` (30 digits). The values used so far by the double evaluator (edge construction above) are off by up to $5 \cdot 10^{-8}$ in absolute terms (10^{-12} of the largest generator; median 10^{-13} relative).

Two technical points: each half of a double-double number must be read back as a double (reading the decimal string at higher precision costs a factor 10^{15}), and the extrapolation weights $n_{k-j}^2/(n_k^2 - n_{k-j}^2)$ must be exact in double-double (in double they cap the accuracy at 10^{-24}).

5. Into the double evaluator. The quad values replace the start values of the double evaluator. The two systems are two bases of the same 503-dimensional space (weights $0, \ldots, 6$: $1, 5, 24, 75, 113, 261, 24$):

- the **503 system** (double): an orthonormal basis, floating-point matrices with entries $|A| \leq 1$;
- the **586 system** (series, quad): integer matrices. Weights 0–4 and 6 have a true integral basis with the same counts. At weight 5 it keeps a redundant spanning set of 344 functions for 261 dimensions: the 247 generators and 97 derivatives of the weight-6 generators. A true integral basis there would need huge integers.

The 503 states are exact rational combinations of the 586 functions (`quad_p0_to_double.py`). With the quad start values, the two evaluations of the error estimate share their start, so they differ only by the path. The estimate adds a term $\epsilon \sum |\text{terms}|$ for the systematic error of the double system, which is the same on every path.

Production regions

Each production region has its own triangle (X, Y) and hub $(0.3, 0.3)$ (`notes/continuation_notes.tex`). Its start values at the hub are obtained in quad as in the decay, but without a series. We integrate

from the decay values at P_0 to the physical image of the hub, along a complex-deformed straight path. On it the two variables that become negative cross zero with a positive imaginary part, which is the Feynman prescription; the root arrives at $-\sqrt{|xyz|}$.

- Two such paths agree to 10^{-27} in all three regions.
- The continuation through the corner, used before, agrees with them to 10^{-11}: an independent check of the continuation.
- The region's integer system ($M'_j = \sum_k c_{kj} M_k$) is integrated from the hub to another point of the region. It agrees with a direct path from P_0 to 10^{-27}.